Curated Research Articles

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- **Electric-Field-Oriented Interfacial Solvent Engineering Enables High-Voltage Ether Electrolytes for Ni-Rich Lithium-Ion Batteries** — score: 1.000 This article presents a novel electric-field-oriented interfacial solvent engineering approach that enhances the stability of ether electrolytes in high-voltage, nickel-rich lithium-ion batteries. By using the solvent 4-cyano-tetrahydropyran, the study shows effective suppression of oxidative decomposition through the creation of a protective layer that shields vulnerable bonds, achieving remarkable oxidative stability up to 4.8 V and demonstrating significant cycling performance with 83.5% capacity retention after 600 cycles. Journal: *Journal of the American Chemical Society ASAP*
- **Designing Polyoxotungstate-Supported Dual-Atom Catalysts for Water Splitting and Metal–Air Batteries: A DFT Investigation** — score: 1.000 This study employs density functional theory (DFT) to evaluate 23 homonuclear dual-transition-metal catalysts supported on polyoxotungstate (POT) for their effectiveness in water splitting and metal–air batteries. The findings reveal that V /POT shows optimal hydrogen adsorption for the hydrogen evolution reaction (HER), while Au /POT and Ni /POT demonstrate strong bifunctional performance in the oxygen evolution reaction (OER) and oxygen reduction reaction (ORR), highlighting the importance of dual-metal configurations in enhancing catalyst performance and understanding their underlying mechanisms through structure-property-activity relationships. Journal: *Energy & Fuels ASAP*
- **Plating-Induced Stress Heterogeneity Dictates Zinc Dendrite Growth** — score: 1.000 The study investigates how plating-induced stress affects the growth of zinc dendrites on metal anodes in zinc-ion batteries, highlighting the correlation between stress heterogeneity and zinc deposition patterns. By introducing a stress-relieving additive, researchers achieved a uniform stress distribution that significantly decreased dendrite formation, leading to improved cycling life and stability of the zinc anode in battery systems. Journal: *Journal of the American Chemical Society ASAP*
- **Dynamic Solvent Participation Drives Sodium Storage in Oxygen Engineered Hard Carbon** — score: 1.000 This study explores sodium storage mechanisms in hard carbon by utilizing citric acid to engineer oxygen on the surface of cellulose-derived hard carbon. The optimized material exhibits a high reversible capacity and highlights the crucial role of solvent interactions during sodium uptake and release, confirming the "adsorption–closed pore filling" pathway and establishing surface modifications as a viable approach to enhance anode performance. Journal: *ACS Energy Letters ASAP*
- **Beyond Phase Selectivity in Conversion Copper Selenides Cathode through Activated Anionic Redox** — score: 1.000 The article discusses an innovative approach to enhancing the performance of copper selenide electrodes in zinc batteries by activating anionic redox processes. This method allows for a significant increase in capacity from 4 mAh g–1 to 413 mAh g–1 and extends lifecycle to 4000 cycles, overcoming limitations imposed by strict phase selectivity. The findings suggest potential for revitalizing the use of copper selenides and similar materials in battery technology. Journal: *Journal of the American Chemical Society ASAP*
- **High-Entropy-Stabilized Air- and Moisture-Resilient Sodium-Ion Cathode: An Experimental and Theoretical Approach** — score: 1.000 The study presents a high-entropy sodium-ion cathode, Na0.98Ca0.02Ni1/7Cu1/7Al1/7Co1/7Fe1/7Mn1/7Ti1/7O2 (TM-7 HEO), which enhances resilience to air and moisture through cation disorder and structural modifications. Experimental findings indicate a reversible capacity of 132 mAh g–1 and sustained performance post exposure to environmental conditions, while theoretical analyses confirm the mechanisms behind its improved stability and electrochemical properties, offering insights for future sodium-ion battery development. Journal: *ACS Applied Energy Materials ASAP*
- **Insights Into Mg2+-Driven Lattice Stabilization and Na+ Kinetics Regulation in O3-Type NaMn0.5Fe0.5O2 Cathodes for Sodium-Ion Batteries** — score: 1.000 The incorporation of Mg2+ into O3-type NaMn0.5Fe0.5O2 cathodes significantly enhances structural stability and Na+

transport kinetics in sodium-ion batteries. The Mg4-NFM composition demonstrates improved electrochemical performance, achieving a discharge capacity of 83.3 mAh g−1 with excellent retention over 100 cycles, as a result of enhanced electronic conductivity and favorable ion transport characteristics due to Mg doping. Journal: *Wiley: Small Methods: Table of Contents*

- **Integrated d-p Orbital Coupling Within High-Entropy Oxide/Halloysite Accelerates Sulfur Reduction Kinetics in Sodium-Sulfur Batteries** — score: 1.000 The article describes the development of a high-entropy oxide/halloysite nanotube hybrid composite that effectively addresses issues in sodium-sulfur batteries, such as polysulfide shuttling and slow conversion kinetics. By integrating d-p orbital coupling and leveraging the structural properties of halloysite, this composite enhances sulfur reduction kinetics and significantly improves battery performance, achieving a high capacity retention over extended cycles, thus presenting a promising solution for more efficient sodium-sulfur batteries. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Correlation between Fe/Mn/Ni Ratio and Structural Properties in P2-Na 2/3 Mn 2/3 Fe 1/3− x Ni x O 2 (x = 0, 0.02, 0.05, 0.1, and 1/3)** — score: 1.000 The study investigates the impact of varying the Fe/Mn/Ni ratio on the structural properties of P2-type layered oxides used in sodium-ion batteries. It reveals that the presence of Ni decreases the Mn^3+ concentration, thereby stabilizing the P2 phase and reducing undesired phase transitions during sodiation/desodiation, especially at Ni concentrations above 0.05. These results demonstrate the critical role of cationic ratios in enhancing the structural integrity of cathode materials. Journal: *ACS Applied Energy Materials ASAP*
- **Gel Polymer Electrolytes for Next-Generation Lithium Batteries: A Review of Interface Engineering and In Situ Polymerization** — score: 1.000 The article reviews gel polymer electrolytes (GPEs) as a promising alternative for rechargeable lithium batteries, highlighting their benefits in ionic conductivity, mechanical stability, and safety. It focuses on the innovative use of in situ polymerization for enhancing GPE performance through improved interfacial compatibility and electrochemical stability. The review further addresses challenges like low-temperature performance and electrolyte decomposition, while emphasizing advanced characterization techniques to understand interphase dynamics. Journal: *Energy & Fuels ASAP*
- **Improving the Calendar Life of Silicon Anodes with Inorganic Solid–Electrolyte Interphases** — score: 1.000 The study evaluates the calendar life of silicon anodes in lithium-ion batteries, revealing that mixed-salt locally superconcentrated electrolytes (LSCEs) significantly outperform traditional carbonate electrolytes in maintaining capacity over time due to their formation of a more stable inorganic solid-electrolyte interphase (SEI). Despite the initial advantages, all cell types showed SEI perturbations after five months, highlighting a need for future electrolyte designs that enhance SEI stability through dynamic reinforcement methods. Journal: *ACS Energy Letters ASAP*
- **Operando Electrochemical Fluorescence Microscopy Highlights Local Buffering and Mass Transport Effects in Polymer-Promoted Electrocatalysis** — score: 0.900 The study introduces a novel operando electrochemical fluorescence microscopy technique to investigate how protic polymer coatings, specifically polyethylenimine (PEI), influence local pH during electrocatalytic hydrogen evolution. Findings reveal that these coatings enhance mass transport of proton donors and provide local buffering, which significantly affects the reaction kinetics, offering insights crucial for designing more effective polymer coatings in proton-coupled electron transfer (PCET) processes. Journal: *Journal of the American Chemical Society ASAP*
- **A Universal Class of Cubic Crystalline Alkali Organic–Inorganic Glyme-Based Halides** — score: 0.900 This article reports a novel class of cubic crystalline alkali glyme-based halides, formed through a self-assembly reaction with halide solid electrolytes. The study explores the structural properties, ionic conductivity, and electrochemical behavior of these materials, revealing their potential for applications in solid-state ionics, particularly in potassium-ion cells where they exhibit enhanced ionic conduction in liquid states. Journal: *Journal of the American Chemical Society ASAP*
- **Three-Dimensional Nanopore -V 2 O 5 Architectures Enabling Stable, Long-Life, High-Power Sodium-Ion Microbatteries** — score: 0.900 This study presents a novel sodium-ion micro-

battery utilizing a three-dimensional coaxial -V2O5/Pt nanotube architecture embedded in anodic aluminum oxide nanopores. This design significantly improves ionic transport and charge-transfer kinetics, resulting in a high areal capacity, excellent rate capability, and impressive cycling stability, suggesting new approaches for optimizing sodium-ion battery performance through nanoscale engineering. Journal: *ACS Applied Energy Materials ASAP*

- **Redox-Programmed Bipolar Membranes from Electronically Wired Ion-Insertion Hosts for Selective Ion Transport** — score: 0.900 The article presents a novel approach using electronically wired ion-insertion hosts as electrochemical membranes to achieve selective ion transport, specifically targeting lithium ion recovery from sodium-rich brine solutions. By coupling lithium-ion insertion and release reactions through a composite electronic–ionic conductor, the study demonstrates a high lithium/sodium separation factor of approximately 2000, highlighting the potential for programmable selectivity through tailored insertion materials. Journal: *Journal of the American Chemical Society ASAP*
- **Interfacial Resistance and Degradation Behavior of Uncoated Single-Crystal-like and Polycrystalline NCM523 in All-Solid-State Batteries Analyzed by Three-Electrode Cells** — score: 0.900 This study investigates the performance and interfacial resistance of uncoated single-crystal-like and polycrystalline NCM523 electrodes in sulfide-based all-solid-state batteries. Using three-electrode cells, the researchers identified distinct resistive components, revealing that single-crystal electrodes exhibit superior cycling stability due to lower charge-transfer and interparticle contact resistance, while polycrystalline electrodes experience capacity loss primarily from increased charge-transfer resistance. Journal: *ACS Applied Energy Materials ASAP*
- **Ni-MOFs Mediated Ni2+/4+ Balancing and Enhanced Metal–Oxygen Covalency Unlocks Fast and Stable Sodium Storage in High-Entropy Layered Cathode** — score: 0.900 The article discusses the use of Ni-based metal-organic frameworks (Ni-MOFs) to create high-entropy O3-type layered oxide cathodes for sodium storage, enhancing their electrochemical performance. By controlling nickel content and strengthening metal-oxygen covalency, these cathodes achieve higher discharge capacity and improved stability, showcasing an initial efficiency of 92.6% and reliable cycling under varied conditions. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Air Sensitivity and Surface Residual Alkali in Sodium-Ion Layered Oxide Cathodes: A Review** — score: 0.800 This review investigates the challenges posed by surface residual alkali in sodium-ion layered oxide cathodes, focusing on the issues of air sensitivity and degradation mechanisms unique to sodium. It outlines potential mitigation strategies, including surface coatings and bulk engineering, while emphasizing the need for further research to enhance the electrochemical performance and industrial viability of these materials. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Isoreticular Covalent Organic Frameworks with Complementary Redox Cores Enable High-Voltage All-COF Sodium Dual-Ion Batteries** — score: 0.800 This article discusses the development of innovative covalent organic frameworks (COFs) designed to enhance sodium dual-ion batteries by using complementary n-type and p-type electrodes. By employing an isoreticular redox-core tailoring strategy, researchers created two crystalline COFs—one based on electron-deficient triazine and the other on electron-rich triphenylamine—yielding a high-voltage battery system capable of operating between 2.0 and 4.2 V, with impressive cycling stability and capacity. Journal: *Journal of the American Chemical Society ASAP*
- **Stage-Controlled Synthesis of MXene/MAX Heterostructures With Local Atomic Disorder for Enhanced Lithium-Ion Storage** — score: 0.800 This study presents a method for synthesizing intermediate MXene/MAX heterostructures through controlled molten-salt etching, highlighting the structural evolution and its impact on lithium-ion storage capabilities. By adjusting etching times, various stages of Nb2GaC to Nb2CTx transformation are characterized, resulting in enhanced lithium storage performance with a notable reversible capacity and excellent cycling stability, demonstrating the potential for engineering MXene-based materials with optimized atomic arrangements for energy applications. Journal: *Wiley: Small Methods: Table of Contents*

- **Janus Ag Nanoskin-Armored Polyimide Bilayer Separator Enables Efficient Li-Nucleation Regulation for Dendrite-Free Lithium Metal Battery** — score: 0.800 The study presents a Janus Ag nanoskin-armored polyimide bilayer separator designed to enhance lithium nucleation regulation in lithium metal batteries, addressing issues of dendrite formation. By facilitating a uniform Li+ flux and reducing overpotential, the separator achieves high Coulombic efficiency and remarkable cycling stability, supporting its potential for secure and efficient lithium metal battery applications. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Medium-Concentration Electrolytes Balance Hydrogen-Bond Disruption and Ion–Ion Interactions for Wide-Temperature Aqueous Trivalent Indium Metal Batteries** — score: 0.800 The study presents a medium-concentration chloride electrolyte that enhances the performance of aqueous trivalent indium metal batteries over a wide temperature range from -70 to +65°C. By adjusting the electrolyte's salt concentration, the research effectively disrupts water's hydrogen bonding while promoting chloride coordination with In3+, achieving excellent anti-freezing properties and reduced side reactions, leading to highly reversible battery operation with remarkable Coulombic efficiencies. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **MOF-Superstructure-Derived 2D Fe2O3/Fe-N-C Nanosheets as Catalysts for Zn-Air Batteries: Synergistically Enhanced Mass Transport and Electrocatalytic Activity** — score: 0.800 The article presents a novel catalyst derived from metal-organic frameworks (MOFs) that features 2D Fe2O3/Fe-N-C nanosheets, designed to enhance the oxygen reduction reaction (ORR) in zinc-air batteries. By improving atomically dispersed iron and utilizing a dual Fe-source approach, the catalyst achieves superior ORR activity and battery performance through optimized mass transport and reduced intermediate binding strength. Journal: *Wiley: Small Methods: Table of Contents*
- **Uniform and Dense Electrodeposited Magnesium Foil Induced by Cationic Surfactant Additive as an Ultra-Thin and High-Utilization Negative Electrode for Rechargeable Magnesium Batteries** — score: 0.800 The study investigates the use of cetyltrimethylammonium bromide as an additive in the electrodeposition of magnesium foils for rechargeable magnesium batteries. This method results in ultrathin, dense, and uniform magnesium electrodes with significantly improved cycling stability and a utilization rate of 83.3%, highlighting its potential for enhancing the performance of high-energy-density batteries. Journal: *Wiley: Small Methods: Table of Contents*
- **Unlocking Three-Dimensional Electron and Ion Transport in High-Loading and Dense Lithium-Ion Battery Electrodes Assisted via Porous Graphene Networks** — score: 0.800 The article discusses the introduction of graphene mesosponge (GMS) as a promising conductive additive for lithium-ion battery electrodes, which enhances three-dimensional electron and ion transport. By incorporating a small amount of GMS in high-loading electrodes, the study shows improved electronic and ionic conductivity, achieving high reversible capacity without compromising the structural integrity during compression. This innovation allows for better energy and power densities, addressing common challenges in battery performance under high-density conditions. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Catalytic Activity Matching Governs Scaffold–Coating Competition in Silicon Anodes** — score: 0.800 The study investigates the relationship between scaffold thickness and coating uniformity in silicon anodes, highlighting that a moderate catalytic activity of the CaSO4 template leads to better performance than a more active MgSO4 template. By balancing scaffold rigidity and coating quality, the CaSO4 template results in a compliant scaffold minimizing lithiation stress and achieving a notable cycle life and efficiency, thus establishing "catalytic activity matching" as a critical design principle in silicon anode fabrication. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Macromolecular Design of Bipolar Donor–Acceptor Polymer Cathodes for Fast and Durable Aluminum Dual-Ion Batteries** — score: 0.800 The article presents the development of four phenothiazine–anthraquinone donor–acceptor polymers that enhance the performance of aluminum dual-ion batteries (ADIBs) by optimizing their macromolecular structure. The resulting polymer, PTZAQ-C3, exhibits impressive electrochemical properties, including a reversible capacity of 243 mAh g−1, rapid charge transport at high current densities, and exceptional durability over 90,000

cycles, demonstrating the effectiveness of structural design in advancing battery technology. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **In Situ Formation of a Hybrid Alloy-Fluoride Interphase for Stable High-Capacity Phosphorus Anodes** — score: 0.800 This study presents the development of a hybrid solid electrolyte interphase (SEI) on phosphorus-carbon anodes through SbF3 doping, which enhances mechanical strength and ionic conductivity. As a result, the anode demonstrates remarkable cycling stability with a capacity retention of 970.10 mAh g−1 after 500 cycles and maintains high-rate performance, addressing the challenges of volume expansion and unstable SEI chemistry in high-capacity lithium-ion batteries. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Reversible Sn4+/Sn0 anode for high-energy-density flow batteries** — score: 0.800 Research has demonstrated that using iodide-mediated intermediates allows for the reversible conversion of Sn4+ to Sn2+ in tin-based aqueous flow batteries. This advancement addresses previous limitations in electrochemistry, leading to enhanced anode energy density and improved long-term battery performance with a potential for four-electron redox reactions. Journal: *Nature Chemistry*
- **Graphene-Like-Armored Nanosheet Assemblies toward Fast and Stable High-Mass-Loading Sodium Storage** — score: 0.800 The article presents a novel anode design for sodium-ion batteries utilizing double-shelled TiO2 spheres coated with a graphene-like layer, which enhances sodium storage efficiency and stability. This design achieves a significant areal capacity and maintains over 90% of its capacity after 3,000 cycles at high mass loading, effectively addressing issues of rate capability and cycle lifespan in energy storage applications. Journal: *Journal of the American Chemical Society ASAP*
- **Mn-Rich Olivine Cathode with Synergistic Lattice Strain Mitigation via Solid-Solution Pathway and Local-Structure Modulation for High-Rate and Stable Lithium Storage** — score: 0.800 The article discusses the development of a manganese-rich olivine cathode designed to enhance lithium storage capabilities through a novel solid-solution pathway. By strategically mitigating lattice strain and optimizing local structures, the cathode demonstrates improved rate performance and stability, making it a promising candidate for energy storage applications. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Pioneering MOF-303 in Lithium–Sulfur Batteries: Coordination Defect Engineering Enables Accelerated Sulfur Redox and Dendrite-Free Plating** — score: 0.800 The study presents a modified separator using defect-engineered MOF-303 (D-MOF-303) for lithium-sulfur batteries, which addresses the shuttle effect of lithium polysulfides and prevents dendrite formation. The introduction of coordination defects enhances sulfur redox reactions and stabilizes lithium plating, leading to significant improvements in cycling performance, rate capability, and durability, thus highlighting the potential of this approach for next-generation energy storage solutions. Journal: *ACS Energy Letters ASAP*
- **Interplay of Composition, Microstructure, and Ionic Conductivity in NaSICON-Type Solid Electrolytes from Scalable Spray-Flame Synthesis** — score: 0.800 This article discusses the development of NaSICON-type solid electrolytes for sodium batteries using a scalable spray-flame synthesis method. The researchers demonstrate that this technique allows for the creation of a nanostructured precursor with enhanced ionic conductivity, revealing that factors such as composition, microstructure, and grain-boundary chemistry collectively influence the material's performance in energy storage applications. Journal: *ACS Applied Energy Materials ASAP*
- **Fabrication and Lithium Storage Performance of Trigonal W 2 N Thin-Film Anode** — score: 0.700 This study outlines the synthesis and electrochemical evaluation of trigonal W N thin-film anodes for lithium-ion batteries. Characterized by high specific capacity and stable performance across multiple cycles, the material exhibits exceptional lithium ion transport dynamics, enabling efficient intercalation and deintercalation processes. The findings suggest that trigonal W N could be a viable candidate for future high-performance battery applications. Journal: *Energy & Fuels ASAP*
- **Interfacial pH Dynamics in Lattice-Nitrogen Protonation to Ammonia** — score: 0.700 This

study introduces a novel method using in situ ATR-FTIR spectroscopy to analyze interfacial pH dynamics during lattice-nitrogen protonation to ammonia in Ti NT MNene catalysts. The findings reveal that while neutral electrolytes enhance ammonia production, the interfacial pH can become alkaline due to cathodic polarization, with the electrolyte's pH significantly impacting the protonation mechanism. The research highlights the importance of coordinating electrolyte pH and current density to optimize ammonia synthesis. Journal: *Journal of the American Chemical Society ASAP*

- **Calendering-Dependent Electrochemical and Cost-Energy Tradeoffs in Layered – Olivine Blend Cathodes** — score: 0.700 This study explores the performance and cost-energy tradeoffs of blended cathodes composed of LiNi0.9Mn0.05Co0.05O2 (NMC) and LiMn0.7Fe0.3PO4 (LMFP), revealing that while blends help mitigate voltage decay and enhance energy retention, they compromise cyclability, especially under heavy calendering. Optimization through niobium-stabilized NMC is shown to maintain rate capability and improve durability, suggesting strategies to balance microstructural effects with cost and energy efficiency in lithium-ion batteries. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Spin State Engineering Coupled with Local Bonding Reconstruction Achieves Wide-Temperature-Range and Long-Lifespan Na4Fe3(PO4)2(P2O7) Cathode** — score: 0.600 The article discusses advancements in engineering the spin state of Na4Fe3(PO4)2(P2O7) cathodes, emphasizing improvements in longevity and performance across a wide temperature range. The approach combines local bonding reconstruction with spin state manipulation, potentially enhancing the efficiency of energy storage devices. Journal: *ScienceDirect Publication: Nano Energy*
- **Unlocking Sequential H+/Mg2+ Co-Intercalation Chemistry in MgMn2O4 for High-Performance Aqueous Magnesium-Ion Batteries** — score: 0.600 The article discusses the innovative co-intercalation chemistry of H+ and Mg2+ in MgMn2O4, which enhances the performance of aqueous magnesium-ion batteries. This research presents a significant advancement in energy storage technology, potentially leading to more efficient and sustainable battery systems. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Upcycling Strategy for Mixed Spent Batteries to Enable NMC622 & Graphite Full Batteries** — score: 0.600 The article presents a novel hydrometallurgical upcycling strategy that effectively transforms mixed spent lithium-ion batteries into high-performance NMC622 cathodes and battery-grade graphite. This integrated approach not only enhances the electrochemical performance and lifespan of new batteries made from recycled materials but also significantly improves economic viability and reduces environmental impact compared to traditional recycling methods. The study highlights the potential for creating a sustainable circular economy in battery production. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Synergistic Tuning of Electrochemical Performance in Ni-Modified CuCo-MOFs for Flexible Energy Storage Devices** — score: 0.500 The article describes the synthesis of a nickel-intercalated bimetallic CuCo-BDC metal-organic framework (MOF) to enhance electrochemical performance for flexible energy storage devices. The study demonstrates that this MOF achieves significant improvements in charge storage, capacity, and durability, making it a promising candidate for application in portable and wearable electronics, with impressive mechanical flexibility and stability over 5000 cycles. Journal: *Energy & Fuels ASAP*
- **Minireview and Perspective of Emerging Metal Cyanamide Architectures for Catalytic and Energy Storage Applications** — score: 0.500 This minireview explores the potential of metal cyanamide materials, emphasizing their unique electronic and structural properties compared to traditional metal oxides and sulfides. It discusses recent advancements in their synthesis and application in catalysis and energy storage, highlighting their dynamic bonding capacities that enable multifunctionality while addressing challenges for future development in material design. Journal: *Energy & Fuels ASAP*
- **Ag Reinforcement in Pd–Ag/MnO 2 Nanoensembles Boosts the Electrocatalytic Performance in Low-Temperature Direct Ethanol Fuel Cells: Studies on Ethanol Oxidation and**

Oxygen Reduction Reaction — score: 0.400 The study examines the Pd–Ag catalyst nanoparticles supported on MnO for use in low-temperature direct ethanol fuel cells, focusing on their enhanced electrocatalytic performance in ethanol oxidation and oxygen reduction reactions. The incorporation of silver allows for improved d-band shifts and increases the electrochemical surface area, resulting in substantial current density output and stability, outperforming traditional Pt/C and Pd/C catalysts. The findings highlight the potential of Pd Ag /MnO as an efficient electrode material, achieving notable power densities and product yields during fuel cell operation. Journal: *ACS Applied Energy Materials ASAP*

- **Visualizing the Site-Specific Kinetics and Terrace-to-Edge Migration of CO Electrooxidation on a Single Pd Nanocatalyst** — score: 0.400 The study employs in situ electrochemical tip-enhanced Raman spectroscopy to investigate CO electrooxidation on individual palladium (Pd) nanocatalysts, revealing that the reaction initiates at low-coordinated edge sites and involves terrace-to-edge CO migration. This migration is shown to sustain the reaction and affect early kinetics, highlighting the relationship between nanoscale structure and electrocatalytic performance, while advancing techniques for site-specific chemical imaging in catalysis. Journal: *Journal of the American Chemical Society ASAP*
- **Observing Elusive Reaction Intermediates with Raman Spectroscopy via Melt-React-Trap** — score: 0.400 The article introduces a novel technique called Melt-React-Trap (MRT) that employs Raman microscopy to identify and study short-lived reaction intermediates. By combining low-temperature spectroscopy with localized heating, MRT allows researchers to generate and freeze reactive species, revealing dynamic transformations like the formation of polyiodides from iodide oxidation, including the detection of previously unidentified intermediates. This method showcases the potential for constructing detailed spectroscopic analyses of transient chemical processes. Journal: *Journal of the American Chemical Society ASAP*
- **Mapping and Modeling Interfacial pH Dynamics at Enzymatic Microelectrode** — score: 0.400 This study explores the interplay between localized pH dynamics and enzyme activity at an enzyme-modified electrode, specifically focusing on bilirubin oxidase for oxygen reduction. Using a combination of modeling and experimental techniques, the research reveals significant interfacial pH changes during enzymatic reactions, highlighting the importance of pH in optimizing bioelectrochemical interfaces for enhanced performance in energy conversion and sensing applications. The findings provide a quantitative framework for future advancements in the field, especially regarding CO2 and H2 conversion reactions. Journal: *Journal of the American Chemical Society ASAP*
- **Predicting Core-Level X-ray Photoemission Spectra of Oxide Surfaces from First-Principles A Case Study for SnO 2** — score: 0.400 The article discusses a first-principles approach using the Z+1 method to predict X-ray photoemission spectroscopy (XPS) spectra for different SnO2 (110) surfaces, including stoichiometric and reduced variants with various vacancies and adsorbates. The predicted spectra align well with experimental data, suggesting that this method can effectively interpret the complex chemical properties of oxide surfaces and may be applied to other materials. Journal: *Journal of the American Chemical Society ASAP*
- **Mechanical atomic-scale defects enable stable Ni-rich single-crystal cathodes** — score: 0.300 The study explores how introducing mechanical atomic-scale defects in Ni-rich single-crystal cathodes through calendering enhances their long-term stability. These intraparticle defects play a crucial role in improving the performance and reliability of rechargeable batteries. Journal: *Joule*
- **Inhibition Mechanisms of Lithium-Ion Battery Deflagration by the Free Space and Aerogel Wall Surface Inside an Explosion-Proof Power Enclosure** — score: 0.300 This study investigates how the free-space volume and silica aerogel wall linings within explosion-proof enclosures affect the deflagration of thermal runaway products from lithium-ion batteries. Experiments reveal that the free-space volume significantly influences deflagration intensity, with specific volume configurations leading to maximum deflagration pressures and rates. Additionally, aerogel liners can markedly reduce deflagration intensity through mechanisms such as heat exchange and pressure-wave attenuation, informing designs for safer battery enclosures in high-risk environments. Journal: *Energy & Fuels*

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- **Continuous Fabrication of Dendrite-Free Zinc Fiber Anodes for Flexible Batteries (Adv. Energy Mater. 37/2026)** — score: 0.300 The authors present a novel continuous fabrication method for dendrite-free zinc fiber anodes using oxygen-functionalized carbon nanotube fibers, which facilitate even zinc nucleation and lead to a uniform metallic coating. This innovative approach results in long, flexible zinc fibers that enhance the performance of wearable and self-powered battery systems. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Bridging Bulk–Interfacial Chemistry and Regulation Boundaries: Advances and Challenges in Aqueous Zinc-Based Flow Batteries** — score: 0.300 The article discusses the recent advancements and ongoing challenges in developing aqueous zinc-based flow batteries, focusing on the complexities of bulk and interfacial chemistry interactions. The authors highlight the need for understanding these interactions to enhance battery performance and address regulatory issues, which are critical for the technology's practical application in energy storage solutions. Journal: *ScienceDirect Publication: Energy Storage Materials*